

FSH40069

## Hydrogen Peroxide 3%

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                               |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | <b>过氧化氢</b><br><b>Hydrogen Peroxide 3%</b>                                                                                                    |
| <b>Cat No. :</b>                                      | H312-4; H312-500; H312P-4; H312SAM-1; H312SAM-2; H312SAM-3; H324-500; XX32455GAL; XXH31212OGAL; XXH31255GA; XXH312200LI; NC1552940; NC2191795 |
| <b>Synonyms</b>                                       | Hydrogen dioxide; Hyperoxide (USP/Certified)                                                                                                  |
| <b>Supplier</b>                                       | Fisher Scientific Company<br>One Reagent Lane<br>Fair Lawn, NJ 07410<br>Tel: (201) 796-7100                                                   |
| <b>Emergency Telephone Number</b>                     | CHEMTREC®, Inside the USA: 800-424-9300<br>CHEMTREC®, Outside the USA: 001-703-527-3887                                                       |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                             |

### SECTION 2. HAZARD IDENTIFICATION

|                                                    |                            |                       |
|----------------------------------------------------|----------------------------|-----------------------|
| <b>Physical State</b><br>Liquid                    | <b>Appearance</b><br>Clear | <b>Odor</b><br>Slight |
| <b>Emergency Overview</b><br>Sensitivity to light. |                            |                       |

#### Classification of the substance or mixture

#### Label Elements

None required

#### **Physical and Chemical Hazards**

None identified.

#### **Health Hazards**

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### **Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

### SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

**SAFETY DATA SHEET**

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| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Water             | 7732-18-5 | 96 - 97  |
| Hydrogen peroxide | 7722-84-1 | 3        |
| Acetic acid       | 64-19-7   | <1       |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If eye irritation persists: Get medical advice/attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Non-combustible. Containers may explode when heated.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Avoid contact with skin and eyes.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing.

**Storage**

Keep container tightly closed in a dry and well-ventilated place. Keep cool and protect from sunlight. Contents may develop pressure upon prolonged storage. Containers should be vented periodically in order to overcome pressure buildup. Do not store in metal containers.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component         | China                                                   | Taiwan                                   | Thailand    | Hong Kong                                                                              |
|-------------------|---------------------------------------------------------|------------------------------------------|-------------|----------------------------------------------------------------------------------------|
| Hydrogen peroxide | TWA: 1.5 mg/m <sup>3</sup>                              | TWA: 1 ppm<br>TWA: 1.4 mg/m <sup>3</sup> | TWA: 1 ppm  | TWA: 1 ppm<br>TWA: 1.4 mg/m <sup>3</sup>                                               |
| Acetic acid       | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> |

| Component         | ACGIH TLV                   | OSHA PEL                                                                                                 | NIOSH                                                                                                  | The United Kingdom                                                                                             | European Union                                                                                                   |
|-------------------|-----------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Hydrogen peroxide | TWA: 1 ppm                  | (Vacated) TWA: 1 ppm<br>(Vacated) TWA: 1.4 mg/m <sup>3</sup><br>TWA: 1 ppm<br>TWA: 1.4 mg/m <sup>3</sup> | IDLH: 75 ppm<br>TWA: 1 ppm<br>TWA: 1.4 mg/m <sup>3</sup>                                               | STEL: 2 ppm 15 min<br>STEL: 2.8 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 1.4 mg/m <sup>3</sup> 8 hr |                                                                                                                  |
| Acetic acid       | TWA: 10 ppm<br>STEL: 15 ppm | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | IDLH: 50 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> | STEL: 37 mg/m <sup>3</sup><br>STEL: 15 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                         | TWA: 25 mg/m <sup>3</sup> (8h)<br>TWA: 10 ppm (8h)<br>STEL: 50 mg/m <sup>3</sup> (15min)<br>STEL: 20 ppm (15min) |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               |             | (minimum requirement) |

## SAFETY DATA SHEET

## Hydrogen Peroxide 3%

|                                   |                 |        |
|-----------------------------------|-----------------|--------|
| Nitrile rubber<br>Neoprene<br>PVC | recommendations | EN 374 |
|-----------------------------------|-----------------|--------|

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                   |                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>   | Long sleeved clothing                                                                                                                                                                                  |
| <b>Respiratory Protection</b>     | No protective equipment is needed under normal use conditions.                                                                                                                                         |
| <b>Large scale/emergency use</b>  | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particle filter |
| <b>Small scale/Laboratory use</b> | Maintain adequate ventilation                                                                                                                                                                          |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Clear                    |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | Slight                   |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | 0 °C / 32 °F             |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 100 °C / 212 °F          | @ 760mmHg                                |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | 23.3 mmHg @ 30°C         |                                          |
| <b>Vapor Density</b>                           | > 1.00 (Air = 1.0)       | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.00                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | Miscible                 |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Hydrogen peroxide                              | -1.1                     |                                          |
| Acetic acid                                    | -0.2                     |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                   |
|---------------------------------|---------------------------------------------------|
| <b>Stability</b>                | Light sensitive. Decomposes on exposure to light. |
| <b>Hazardous Reactions</b>      | None under normal processing.                     |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.          |
| <b>Conditions to Avoid</b>      | Exposure to light. Heat. Incompatible products.   |
| <b>Materials to avoid</b>       | Metals.                                           |

**Hazardous Decomposition Products** Oxygen.

## SECTION 11. TOXICOLOGICAL INFORMATION

**Product Information**

**(a) acute toxicity;**  
**Toxicology data for the components**

| Component         | LD50 Oral                                                                               | LD50 Dermal            | LC50 Inhalation                           |
|-------------------|-----------------------------------------------------------------------------------------|------------------------|-------------------------------------------|
| Water             | -                                                                                       | -                      | -                                         |
| Hydrogen peroxide | 376 mg/kg ( Rat ) (90%)<br>910 mg/kg ( Rat ) (20-60%)<br>1518 mg/kg ( Rat ) (8-20% sol) | >2000 mg/kg ( Rabbit ) | LC50 = 2000 mg/m <sup>3</sup> ( Rat ) 4 h |
| Acetic acid       | 3310 mg/kg ( Rat )                                                                      | -                      | > 40 mg/L ( Rat ) 4 h                     |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Based on available data, the classification criteria are not met

**(d) respiratory or skin sensitization;**  
**Respiratory** Based on available data, the classification criteria are not met  
**Skin** Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;** Based on available data, the classification criteria are not met  
**Target Organs** None known.

**(j) aspiration hazard;** Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and** No information available

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delayed

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component         | Freshwater Fish                                                                          | Water Flea         | Freshwater Algae  | Microtox                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrogen peroxide | LC50: 16.4 mg/L/96h<br>(P.promelas)                                                      | EC50 7.7 mg/L/24h  | EC50 2.5 mg/L/72h |                                                                                                                                                                                 |
| Acetic acid       | Pimephales promelas:<br>LC50 = 88 mg/L/96h<br>Lepomis macrochirus:<br>LC50 = 75 mg/L/96h | EC50 = 95 mg/L/24h | -                 | Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/15 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/25 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/5 min |

## Persistence and Degradability

Persistence  
Degradability

Persistence is unlikely, based on information available, Miscible with water.  
Not relevant for inorganic substances.

## Bioaccumulative Potential

Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Hydrogen peroxide | -1.1    | No data available             |
| Acetic acid       | -0.2    | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion Potential

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused  
Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

## Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

## Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

Not Regulated

## IMDG/IMO

Not regulated

## IATA

Not regulated

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Japan (ISHL), Japan (ISHL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Water             | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |
| Hydrogen peroxide | X                                                   | X                                       | X    | X     | 231-765-0 | X    | X   | X     | X    | X    | X    | KE-20204 |
| Acetic acid       | X                                                   | X                                       | X    | X     | 200-580-7 | X    | X   | X     | X    | X    | X    | X        |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 09-Jun-2009  
Revision Date 14-May-2024  
Revision Summary SDS sections updated.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

**Hydrogen Peroxide 3%**

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Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**